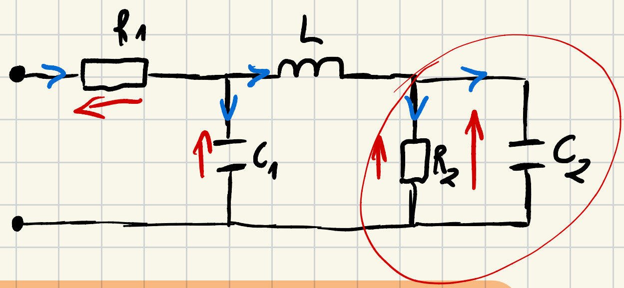
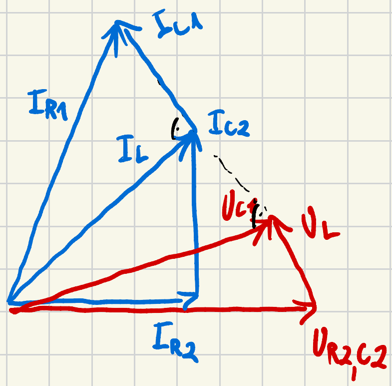
ĆWICZENIA 10

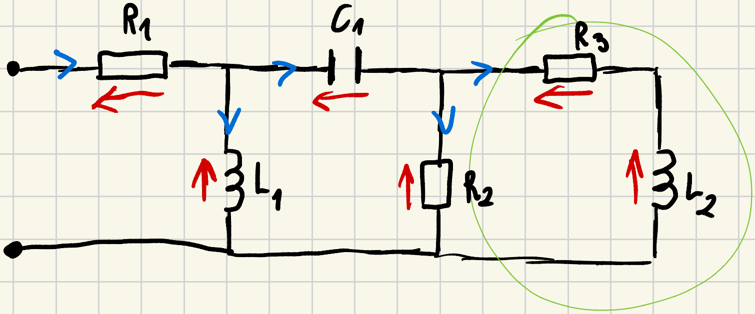
K. 3 Z. 1

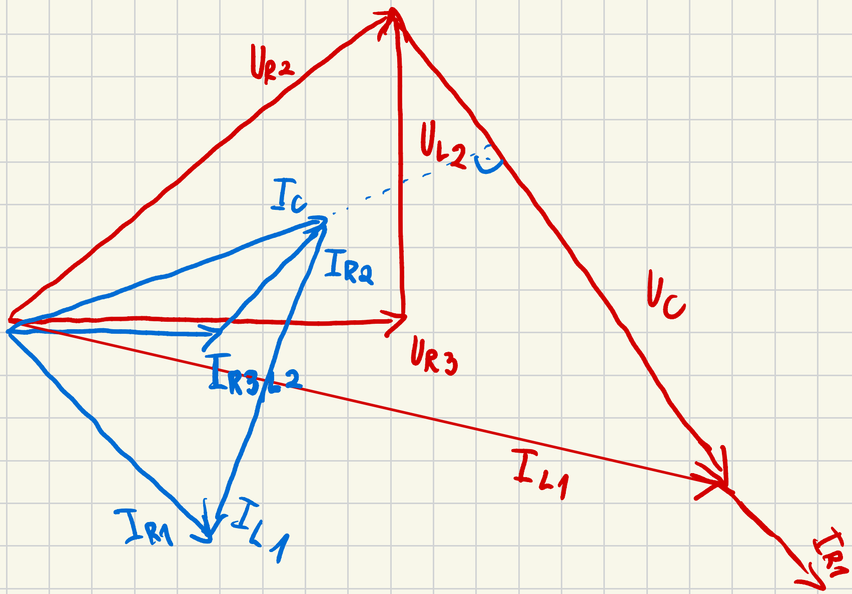


Patrzymy od końca układu

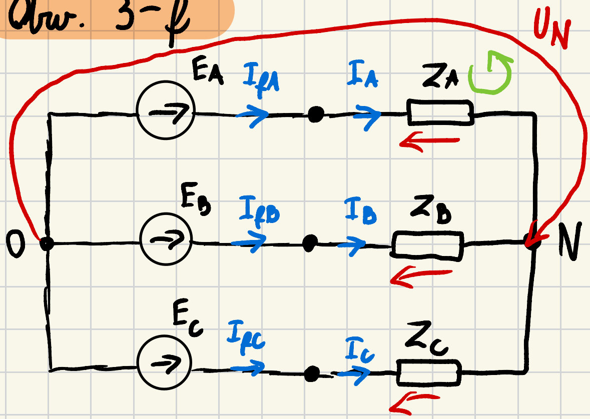


c)



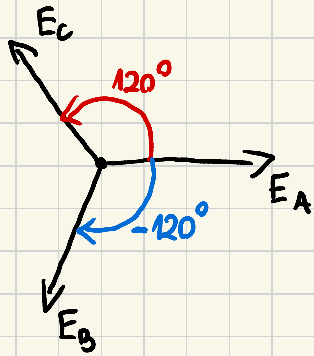


Obw. 3-f



Tu może być inna liczba

$$|E_f| = 200 \text{ [V]}$$



1. Odbiornik 3-f. 3-p. symetryczny

$$Z_A = 10\ \Omega$$

$$Z_B = 10\ \Omega$$

$$Z_C = 10\ \Omega$$

$$Y = Y_A = Y_B = Y_C$$

$$E_A = |E_f| e^{j0^\circ}$$

$$E_B = |E_f| e^{-j120^\circ} = 200\left(-\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}\right)$$

$$E_C = |E_f| e^{j120^\circ} = 200\left(-\tfrac{1}{2} + j\tfrac{\sqrt{3}}{2}\right)$$

$$U_N = \frac{E_A \cdot Y_A + E_B \cdot Y_B + E_C \cdot Y_C}{Y_A + Y_B + Y_C}$$

$$U_N = \frac{Y\underbrace{(E_A + E_B + E_C)}_{=0}}{3Y} = 0$$

Z oczek NPK to wynika

$$E_A - Z_A \cdot I_A - U_N = 0$$

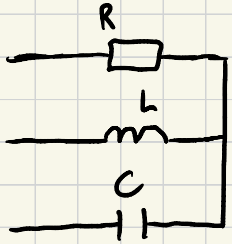
$$I_A = (E_A - \underbrace{U_N}_{=0}) \cdot Y_A = E_A \cdot Y_A$$

Analogicznie

$$I_B = (E_B - U_N) \cdot Y_B = E_B \cdot Y_B$$

$$I_C = (E_C - U_N) \cdot Y_C = E_C \cdot Y_C$$

2. Odbiornik 3-f. 3-p. niesymetryczny



$R = 10\,\Omega \rightarrow Z_R = 10\,\Omega$

$X_L = 10\,\Omega \rightarrow Z_L = j10\,\Omega$

$X_C = 10\,\Omega \rightarrow Z_C = -j10\,\Omega$

$Z_R \neq Z_L \neq Z_C$, więc ten obwód nie jest symetryczny

$$U_N = \frac{E_A \cdot Y_A + E_B \cdot Y_B + E_C \cdot Y_C}{Y_A + Y_B + Y_C}$$

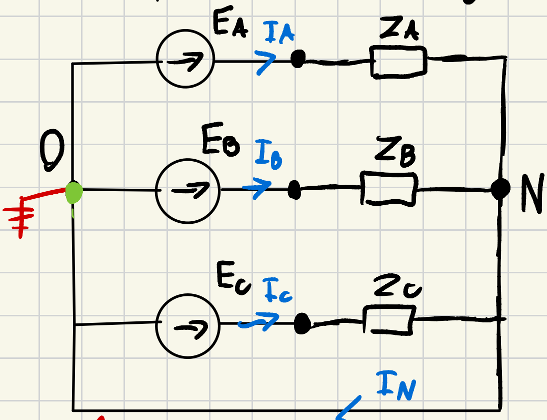
Tu będzie jakaś wartość różna od zera

$$I_A = (E_A - U_N) \cdot Y_A$$

$$I_B = (E_B - U_N) \cdot Y_B$$

$$I_C = (E_C - U_N) \cdot Y_C$$

3. odb., 3-f., 4-p. niesymetryczny



$Z_A \neq Z_B \neq Z_C$

Przez tą pustą gałąź

$U_N = 0$

$$I_A = E_A \cdot Y_A$$

$U_N = 0 \Rightarrow$ $I_B = E_B \cdot Y_A$;

$$I_C = E_C \cdot Y_C$$

$$I_N = I_A + I_B + I_C$$

(PPK dla zielonego węzła)

4. Odb. 3-f 4-p symetryczny

$U_N = 0$

$I_A = E_A \cdot Y_A \quad I_B = E_B \cdot Y_B \quad I_C = E_C \cdot Y_C$

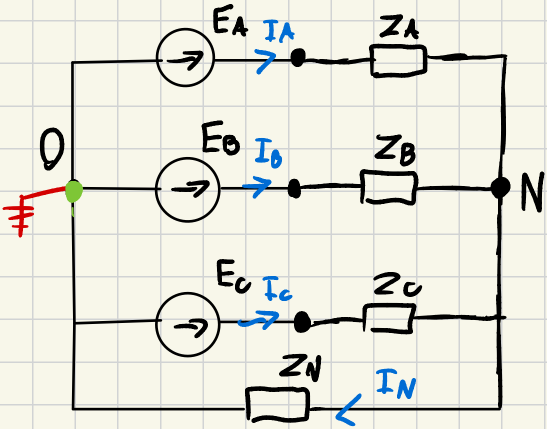
Symetryczny, więc $Y_A = Y_B = Y_C = Y$

$I_A = E_A \cdot Y \quad I_B = E_B \cdot Y \quad I_C = E_C \cdot Y$

$$I_N = I_A + I_B + I_C = (\overbrace{E_A + E_B + E_C}^{=0}) \cdot Y = 0$$

5. Odb. 3-f. 4-p. $Z_N \neq 0$ nsym.

$Z_A \neq Z_B \neq Z_C$

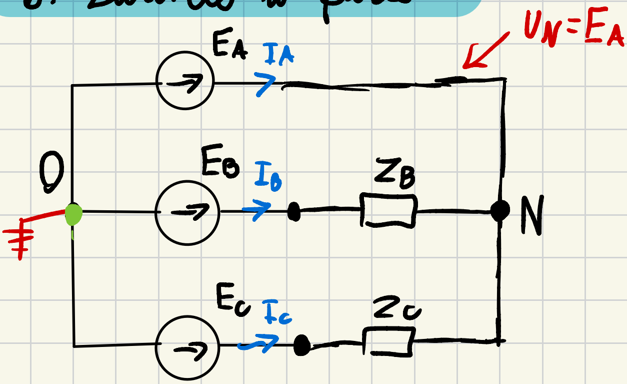


$$U_N = \frac{E_A \cdot Y_A + E_B \cdot Y_B + E_C \cdot Y_C}{Y_A + Y_B + Y_C + Y_N}$$

Jeśli odb. sym. to $U_N = 0$

(nie w tym przypadku, ale OSOWSKI często daje takie zadania, aby $U_N = 0$)

6. Zwarcie w fazie



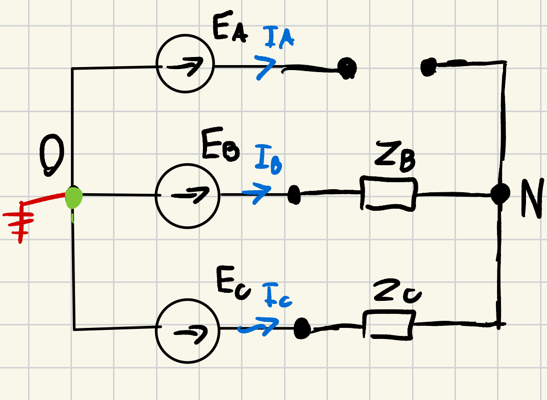
$$U_N = \frac{E_A \cdot \cancel{Y_A} + \cancel{E_B \cdot Y_B} + \cancel{E_C \cdot Y_C}}{\cancel{Y_A} + \cancel{Y_B} + \cancel{Y_C}}$$

$$I_B = (E_B - U_N) \cdot Y_B$$

$$I_C = (E_C - U_N) \cdot Y_C$$

$$I_A = -I_B - I_C$$

7. Przerwa w fazie



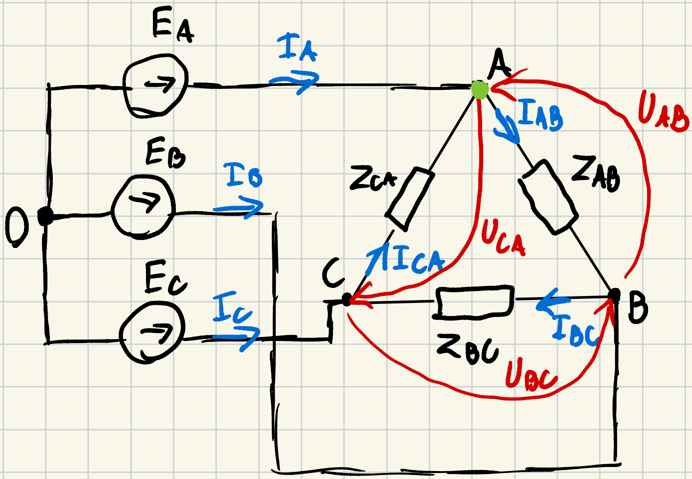
$$U_N = \frac{E_A \cdot 0 + E_B \cdot Y_B + E_C \cdot Y_C}{Y_B + Y_C}$$

$$I_A = 0\,[A]$$

$$I_B = (E_B - U_N) \cdot Y_B$$

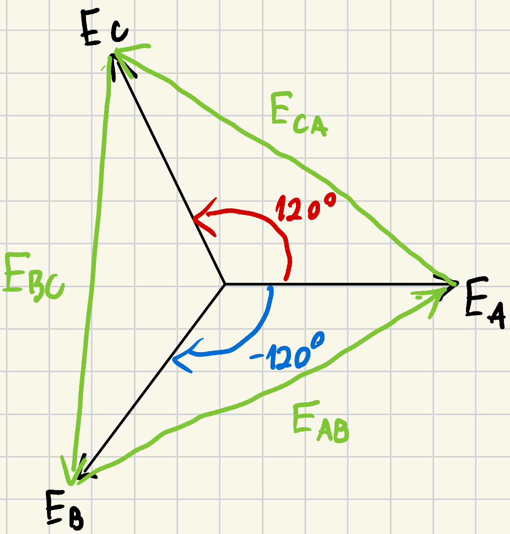
$$I_C = (E_C - U_N) \cdot Y_C$$

8. Odb. 3-p 3-f Δ



Może być inna liczba

$$|E_f| = 200\ [V]$$



$$E_{AB} = \sqrt{3}\,|E_f| \cdot e^{j30^\circ}$$

$$E_{BC} = \sqrt{3}\,|E_f| \cdot e^{-j90^\circ}$$

$$E_{CA} = \sqrt{3} \cdot E_f \cdot e^{j150^\circ}$$

$$E_{AB} = U_{AB}$$

$$I_{AB} = \frac{E_{AB}}{Z_{AB}}$$

$$I_{BC} = \frac{E_{BC}}{Z_{BC}}$$

$$I_{CA} = \frac{E_{CA}}{Z_{CA}}$$

Używam PPK

$$I_A + I_{CA} - I_{AB} = 0$$

$$I_A = I_{AB} - I_{CA}$$

(Analogicznie reszta prądów I_B, I_C)